Multimodal Emotion Recognition using Speech and Text

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Technologies Used

- Python
- PyTorch
- Hugging Face Transformers
- Librosa
- Scikit-Learn

Models Implemented

- MFCC + CNN (Speech Baseline)
- Wav2Vec2 + BiLSTM (Speech Model)
- DistilBERT + BiLSTM (Text Model)
- Speech–Text Fusion Model

Dataset

Toronto Emotional Speech Set (TESS)

Submission

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1. Introduction

Emotion recognition is an important research area within Artificial Intelligence (AI), Speech Processing, and Natural Language Processing (NLP). The objective of emotion recognition systems is to automatically identify human emotional states from different forms of communication such as speech, facial expressions, text, and physiological signals. Accurate emotion recognition can improve human-computer interaction by enabling intelligent systems to understand and respond appropriately to users' emotional conditions.

Speech is one of the most informative modalities for emotion recognition because emotional states influence various acoustic characteristics, including pitch, tone, speaking rate, intensity, and prosody. Modern deep learning models can learn these complex patterns directly from raw audio signals and achieve high performance in speech emotion recognition tasks. Similarly, textual information can also provide clues about emotional states through semantic and contextual meaning. Recent advances in transformer-based language models have significantly improved the ability to extract meaningful representations from text.

Multimodal emotion recognition combines information from multiple modalities, such as speech and text, to improve robustness and performance. By integrating complementary information from different sources, multimodal systems can potentially achieve better emotion classification compared to single-modality approaches. However, the effectiveness of multimodal fusion depends on the quality and relevance of information contained in each modality.

This project investigates emotion recognition using speech, text, and multimodal fusion approaches on the Toronto Emotional Speech Set (TESS) dataset. Multiple deep learning architectures were implemented and evaluated, including an MFCC-based CNN baseline model, a Wav2Vec2-BiLSTM speech model, a DistilBERT-BiLSTM text model, and a multimodal fusion architecture combining speech and text representations. The performance of these models was compared using standard evaluation metrics such as Accuracy, Weighted F1-Score, and Confusion Matrices.

Objectives of the Project

1. Develop a speech-based emotion recognition baseline using MFCC features and a CNN classifier.
2. Implement a speech emotion recognition model using pretrained Wav2Vec2 representations and a BiLSTM classifier.
3. Develop a text-based emotion recognition model using DistilBERT and BiLSTM architecture.
4. Design a multimodal fusion model that combines speech and text representations for emotion classification.

5. Compare the performance of speech, text, and fusion approaches using quantitative evaluation metrics.
6. Analyze model behavior through confusion matrices and error analysis.
7. Study the effectiveness of multimodal fusion for emotion recognition on the TESS dataset.

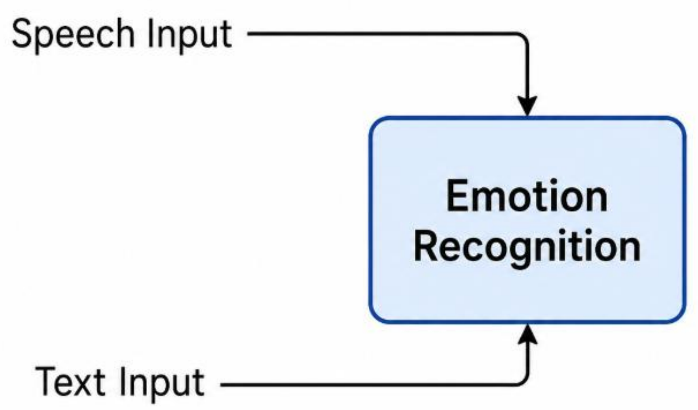


Figure 1: High-level overview of the multimodal emotion recognition system.

2. Dataset Description

2.1 Toronto Emotional Speech Set (TESS)

The Toronto Emotional Speech Set (TESS) is a publicly available emotional speech dataset designed for emotion recognition research. The dataset contains speech recordings produced by two female speakers representing different age groups. Each recording consists of a single English word spoken with a specific emotional expression.

The dataset was developed to facilitate the evaluation of speech emotion recognition systems by providing high-quality recordings with clearly labeled emotional categories. Since the emotions are intentionally expressed by speakers, the dataset provides a controlled environment for studying emotional characteristics in speech.

For this project, TESS was used as the primary dataset for speech, text, and multimodal emotion recognition experiments.

2.2 Speaker Groups

The dataset contains recordings from two female speakers:

Speaker Group	Description
OAF	Older Adult Female
YAF	Younger Adult Female

These speaker groups introduce age-related variability in speech characteristics and provide an opportunity to evaluate model performance across different speaker profiles.

2.3 Emotion Categories

The dataset contains seven emotional classes:

Label	Emotion
0	Angry
1	Disgust
2	Fear
3	Happy
4	Neutral
5	Pleasant Surprise
6	Sad

Each emotion category contains a balanced number of recordings, making the dataset suitable for multi-class classification tasks.

2.4 Dataset Organization

The dataset is organized into emotion-specific folders. Each audio file follows a structured naming convention:

OAF_back_angry.wav
YAF_chair_happy.wav

where:

- The first component identifies the speaker group.
- The second component represents the spoken word.
- The third component represents the emotion label.

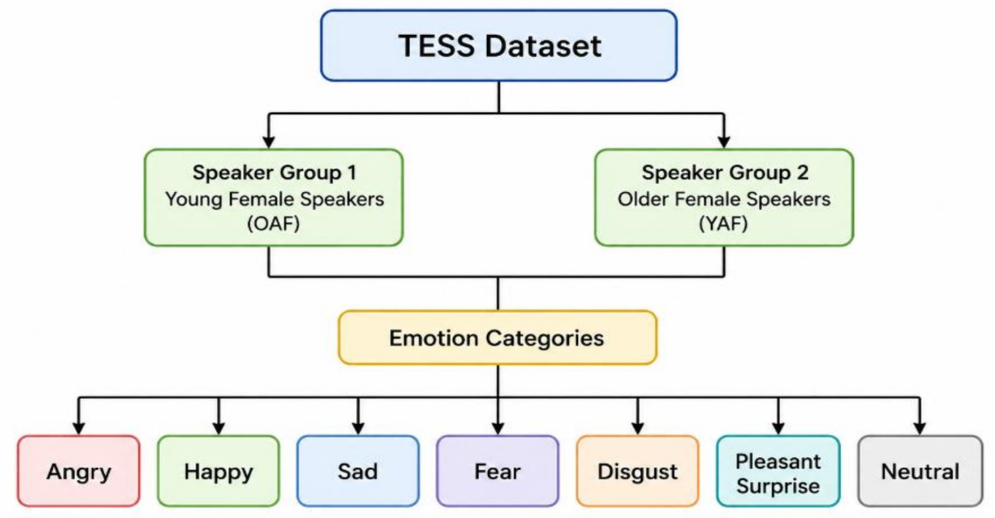


Figure 2: Organization of the TESS dataset showing speaker groups and emotion categories.

2.5 Metadata Generation

To support multiple learning pipelines, a metadata file was generated containing information about every audio sample in the dataset.

The metadata extraction process automatically parsed the file names and created structured records containing:

Column	**Description**
path	Path to audio file
word	Spoken word
emotion	Emotion category
speaker_group	OAF or YAF
label	Numerical emotion label

2.6 Train-Test Split

To ensure unbiased model evaluation, the dataset was divided into training and testing subsets using stratified sampling.

The split ratio was:

Training Set : 80%
Testing Set : 20%

Stratified splitting was used to preserve the class distribution of emotion labels across both subsets.

Dataset statistics:

Subset	Samples
Training	2240
Testing	560
Total	2800

This approach ensures that all emotion categories are equally represented during training and evaluation.

2.7 Data Utilization Across Pipelines

The generated metadata enabled the development of three independent learning pipelines:

Pipeline	Inputs Used
Speech Pipeline	Audio path + emotion label
Text Pipeline	Spoken word + emotion label
Fusion Pipeline	Audio path + spoken word + emotion label

This unified metadata structure simplified experimentation and ensured consistency across all models.

3. Methodology

3.1 Overview

This project investigates emotion recognition using three approaches:

1. Speech-based emotion recognition
2. Text-based emotion recognition
3. Multimodal speech-text fusion

To establish a performance benchmark, a traditional MFCC-based convolutional neural network (CNN) was first implemented as a baseline model. Subsequently, pretrained transformer-based architectures were employed for speech and text processing. Finally, a multimodal fusion model was developed by combining representations extracted from both modalities.

The complete workflow consists of:

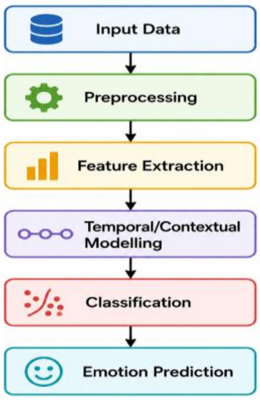
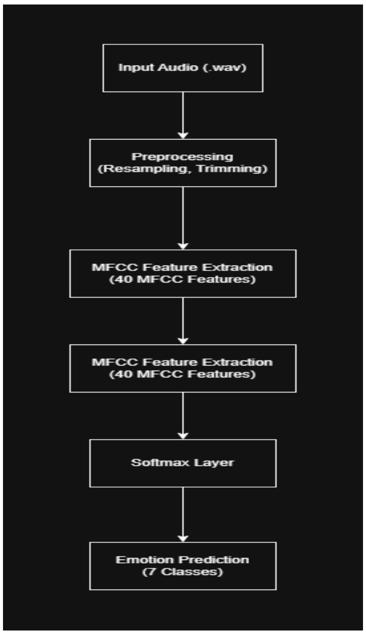


Figure 3: Overall workflow of the proposed multimodal emotion recognition framework from input acquisition to final emotion prediction.

3.2 Speech Baseline Model (MFCC + CNN)

Figure 3: Speech Baseline Architecture



3.2.1 Audio Preprocessing

Each audio sample was loaded using the Librosa library and resampled to a sampling frequency of 16 kHz. Silence segments present at the beginning and end of recordings were removed using silence trimming techniques.

Preprocessing steps:

1. Audio loading
2. Resampling to 16 kHz
3. Silence trimming
4. Feature extraction

These operations help reduce unnecessary variability and improve feature consistency across samples.

3.2.2 MFCC Feature Extraction

Mel Frequency Cepstral Coefficients (MFCCs) were extracted from each audio signal. MFCCs are widely used in speech processing because they provide a compact representation of the spectral properties of speech while approximating human auditory perception.

For this project:

Number of MFCC coefficients = 40

Fixed temporal length = 200 frames

Each audio sample was converted into a feature matrix of size:

40×200

which served as input to the CNN classifier.

3.2.3 CNN Classifier

A convolutional neural network was employed to learn discriminative patterns from the MFCC feature maps.

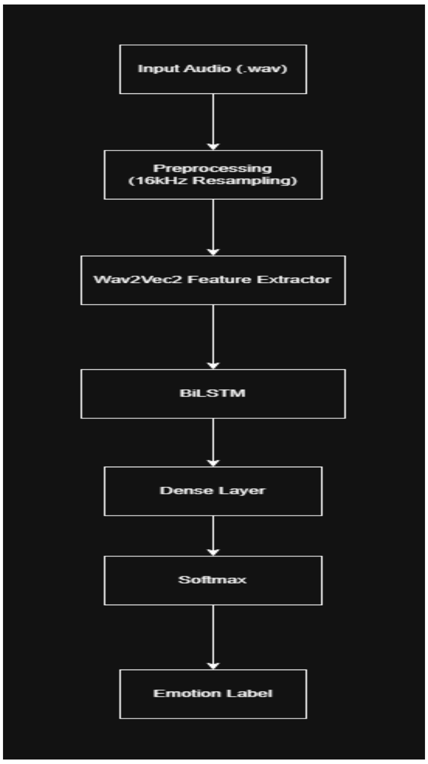
The architecture consisted of:

- Convolution layer
- ReLU activation
- Max pooling
- Convolution layer
- ReLU activation
- Max pooling
- Fully connected layers
- Softmax classifier

The CNN learns local spectral-temporal patterns associated with different emotions.

3.3 Speech Emotion Recognition using Wav2Vec2 + BiLSTM

Figure 4: Wav2Vec2 + BiLSTM Architecture



3.3.1 Motivation

Although MFCC features are effective, they are handcrafted representations and may not capture all emotional characteristics present in speech.

To overcome this limitation, a pretrained Wav2Vec2 model was employed for automatic feature extraction from raw audio waveforms.

3.3.2 Wav2Vec2 Feature Extraction

Wav2Vec2 is a self-supervised speech representation model developed by Meta AI.

The model learns meaningful acoustic representations directly from large amounts of unlabeled speech data and has demonstrated strong performance across numerous speech processing tasks.

In this project:

Model: facebook/wav2vec2-base was used as the pretrained feature extractor.

The input to the model consists of raw audio waveforms sampled at 16 kHz.

3.3.3 Temporal Modelling using BiLSTM

Speech emotions evolve over time and therefore temporal dependencies are important.

To capture sequential information, a Bidirectional Long Short-Term Memory (BiLSTM) network was placed after Wav2Vec2.

Configuration:

Hidden Size = 128

Bidirectional = True

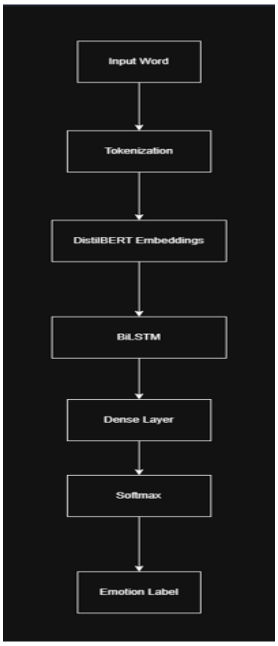
3.3.4 Classification Layer

The final hidden representation produced by the BiLSTM was passed through:

- Fully connected layer
- Dropout layer
- Softmax classifier to predict one of the seven emotion classes.

3.4 Text Emotion Recognition using DistilBERT + BiLSTM

Figure 5: DistilBERT + BiLSTM Architecture



3.4.1 Text Preparation

The textual modality was derived from the spoken word associated with each audio recording.

Examples: back, chair, young, bar, boat

These words were extracted automatically during metadata generation.

3.4.2 Tokenization

Text inputs were processed using the DistilBERT tokenizer.

Tokenization converts each word into numerical token identifiers that can be processed by transformer models.

The tokenizer also generates:

- Input IDs
- Attention Masks which are required by DistilBERT.

3.4.3 DistilBERT Embeddings

DistilBERT is a lightweight transformer language model derived from BERT.

The pretrained model: distilbert-base-uncased was used to generate contextual embeddings for each input word.

3.4.4 Contextual Modelling

Although the dataset contains isolated words rather than complete sentences, a BiLSTM layer was incorporated to maintain architectural consistency and provide contextual modelling capabilities.

The BiLSTM processes transformer embeddings and generates a compact representation for classification.

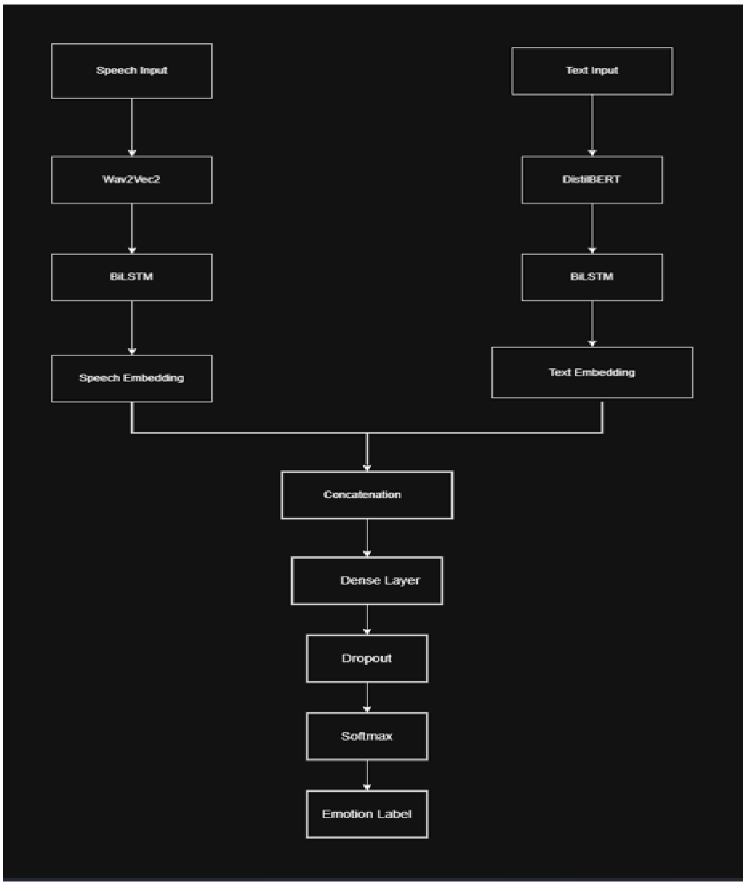
3.4.5 Classification Layer

The BiLSTM output was passed through:

- Fully connected layer
- Dropout layer
- Softmax classifier to predict the emotion label.

3.5 Multimodal Fusion Model

Figure 6: Speech-Text Fusion Architecture



3.5.1 Motivation

Speech and text contain complementary sources of information.

Speech captures:

- Pitch
- Energy
- Prosody
- Speaking rate

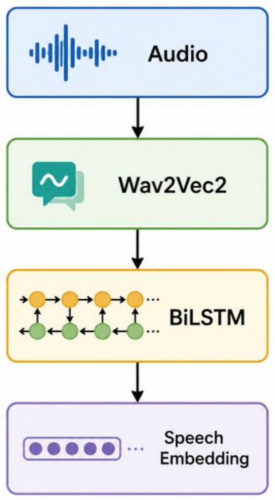
Text captures:

- Semantic content
- Lexical information

Combining both modalities may improve emotion recognition performance.

3.5.2 Speech Branch

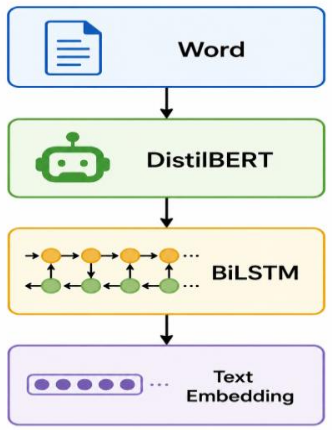
The speech branch consisted of:



A fixed-dimensional speech representation was obtained through temporal pooling of the BiLSTM outputs.

3.5.3 Text Branch

The text branch consisted of:



A fixed-dimensional textual representation was similarly generated.

3.5.4 Feature Fusion

Speech and text embeddings were combined using feature concatenation.

Fused Representation = Speech Embedding + Text Embedding

This operation integrates information from both modalities into a unified feature vector.

3.5.5 Classification

The fused representation was passed through:

- Fully connected layer
- ReLU activation
- Dropout
- Softmax classifier to predict the final emotion category.

To improve training stability on the relatively small TESS dataset, the pretrained Wav2Vec2 and DistilBERT encoders were frozen and only the BiLSTM and classification layers were optimized during training.

4. Experimental Setup

4.1 Development Environment

All models were implemented and evaluated using Python and modern deep learning libraries. Development was carried out in Visual Studio Code, while version control and project management were maintained using Git and GitHub.

The experimental environment consisted of:

Component	Specification
Programming Language	Python 3.12
IDE	Visual Studio Code
Deep Learning Framework	PyTorch
NLP Library	Hugging Face Transformers
Audio Processing	Librosa
Machine Learning Utilities	Scikit-Learn
Visualization	Matplotlib, Seaborn
Version Control	Git & GitHub

4.2 Hardware Configuration

Model training and evaluation were performed on a local machine using CPU-based computation.

Component	Specification
Operating System	Windows
Processing Device	CPU
Memory	16GB
Development Platform	VS Code

Although GPU acceleration can significantly reduce training time for transformer-based architectures, the experiments in this project were successfully completed using CPU resources.

4.3 Data Preparation

The TESS dataset was first converted into a structured metadata format containing:

path

word

emotion

speaker_group

label

The metadata file served as a common source for all pipelines.
The dataset was divided using stratified sampling to maintain equal class distributions across training and testing subsets.

Dataset Split	Samples
Training Set	2240
Testing Set	560
Total Samples	2800

Training samples represented 80% of the dataset, while the remaining 20% were reserved for evaluation.

4.4 Speech Baseline Configuration

The MFCC-based baseline model utilized handcrafted acoustic features extracted from each audio recording.

Feature extraction parameters:

Parameter	Value
Sampling Rate	16 kHz
MFCC Coefficients	40
Fixed Sequence Length	200 Frames

Training configuration:

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	20
Loss Function	CrossEntropyLoss

4.5 Wav2Vec2 Speech Model Configuration

The speech emotion recognition model employed a pretrained Wav2Vec2 encoder followed by a Bidirectional LSTM classifier.

Model configuration:

Parameter	Value
Pretrained Model	facebook/wav2vec2-base
LSTM Hidden Size	128
Bidirectional	Yes
Output Classes	7

Training configuration:

Parameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-5}
Batch Size	20
Epochs	5
Loss Function	CrossEntropyLoss

4.6 Text Model Configuration

The text-based emotion recognition model utilized DistilBERT embeddings followed by a BiLSTM classifier.

Model configuration:

Parameter	Value
Pretrained Model	distilbert-base-uncased
LSTM Hidden Size	128
Bidirectional	Yes
Output Classes	7

Training configuration:

Parameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-5}
Batch Size	20
Epochs	5
Loss Function	CrossEntropyLoss

4.7 Fusion Model Configuration

The multimodal architecture combined speech and text representations through feature-level fusion.

To improve training stability on the relatively small dataset, pretrained encoder parameters were frozen during optimization.

Frozen components:

- Wav2Vec2 Encoder
- DistilBERT Encoder

Trainable components:

- Speech BiLSTM
- Text BiLSTM
- Fusion Classifier

Training configuration:

Parameter	**Value**
Optimizer	AdamW
Learning Rate	1×10^{-4}
Batch Size	20
Epochs	10
Loss Function	CrossEntropyLoss

4.8 Evaluation Metrics

Model performance was evaluated using standard classification metrics.

Accuracy

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Predictions}$$

Higher accuracy indicates better classification performance.

Weighted F1-Score

The weighted F1-score balances precision and recall while accounting for class distribution.

It is particularly useful for evaluating multi-class classification systems and provides a more comprehensive assessment than accuracy alone.

Confusion Matrix

Confusion matrices were generated for all models to visualize classification performance and identify commonly confused emotion categories.

These matrices provide insight into model strengths, weaknesses, and class-specific behavior.

5. Results and Analysis

5.1 Quantitative Results

The performance of all implemented models was evaluated using Accuracy and Weighted F1-Score on the test dataset. The results are summarized in Table 1.

Table 1: Performance Comparison of Implemented Models

Model	Modality	Accuracy (%)	Weighted F1 (%)
MFCC + CNN	Speech	99.82	99.82
Wav2Vec2 + BiLSTM	Speech	**100.00**	**100.00**
DistilBERT + BiLSTM	Text	13.04	3.87
Fusion Model	Speech + Text	89.46	89.16

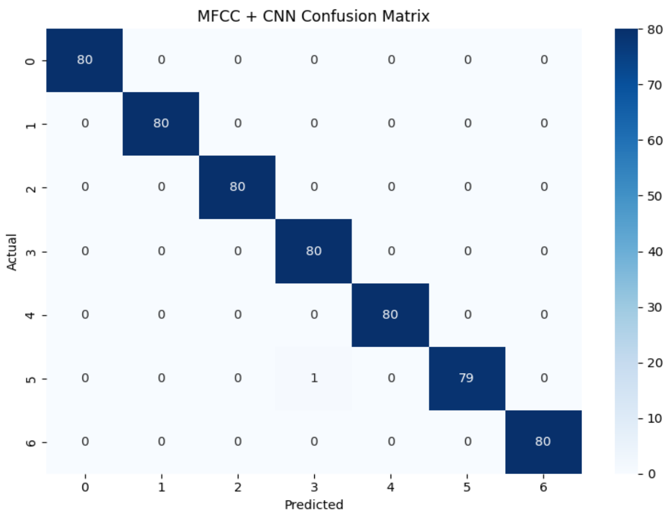
From the results, the Wav2Vec2 + BiLSTM model achieved the highest performance, obtaining perfect classification accuracy on the test set. The MFCC-based CNN baseline also demonstrated excellent performance, indicating that emotional information in the TESS dataset is strongly represented through acoustic characteristics.

The text-only model achieved substantially lower performance compared to speech-based approaches. This result highlights the limited emotional information contained in isolated words within the TESS dataset.

The multimodal fusion model achieved strong overall performance but did not surpass the best speech-only model. This suggests that speech representations already contain highly discriminative emotional information, while the textual modality contributes relatively little additional information.

5.2 Speech Baseline Analysis (MFCC + CNN)

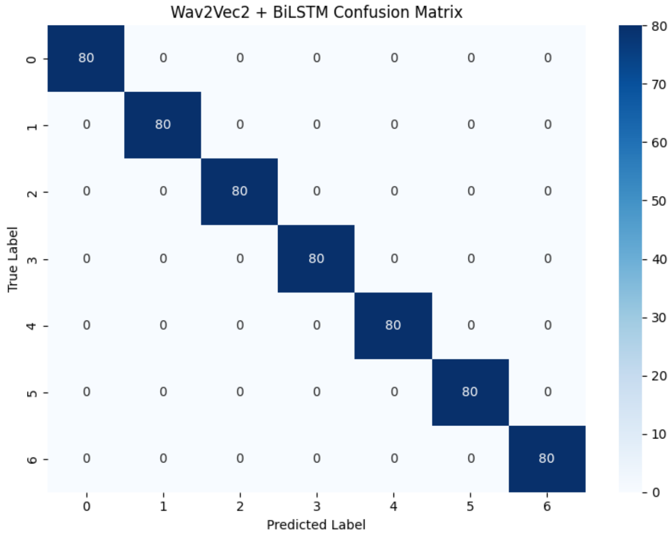
Figure 7: Confusion Matrix for MFCC + CNN Model



The MFCC-based CNN model achieved an accuracy of 99.82%, demonstrating that handcrafted acoustic features remain highly effective for emotion recognition on controlled datasets such as TESS.
Only a very small number of samples were misclassified, indicating that the CNN successfully learned discriminative patterns from MFCC feature maps. The balanced confusion matrix further suggests that the model performed consistently across all seven emotion categories.

5.3 Wav2Vec2 + BiLSTM Analysis

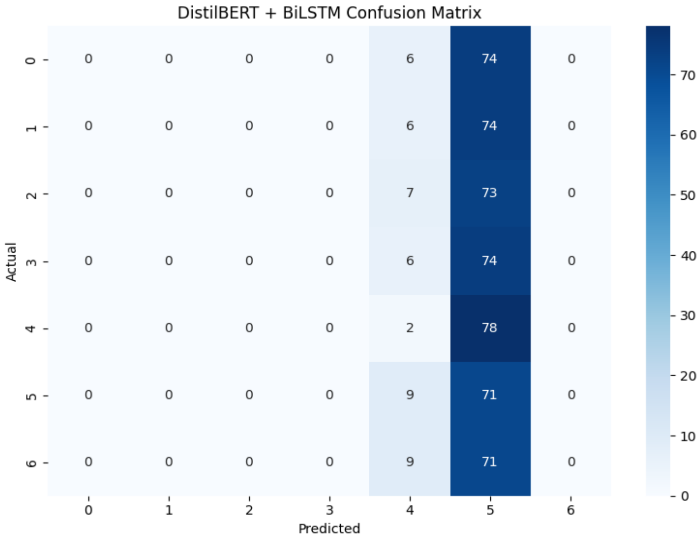
Figure 8: Confusion Matrix for Wav2Vec2 + BiLSTM Model



The Wav2Vec2 + BiLSTM architecture achieved perfect classification performance with:
Accuracy = 100.00%
Weighted F1 = 100.00%
The confusion matrix shows that every test sample was correctly classified, with no observed misclassifications.
This result demonstrates the effectiveness of pretrained speech representations learned through self-supervised training. Unlike handcrafted MFCC features, Wav2Vec2 automatically extracts rich acoustic representations that capture subtle emotional characteristics present in speech signals.
The BiLSTM layer further enhances performance by modelling temporal dependencies and emotional progression within speech sequences.
Overall, the Wav2Vec2 model produced the strongest performance among all evaluated approaches.

5.4 Text Model Analysis

Figure 9: Confusion Matrix for DistilBERT + BiLSTM Model



The text-based model achieved:

Accuracy = 13.04%

Weighted F1 = 3.87%

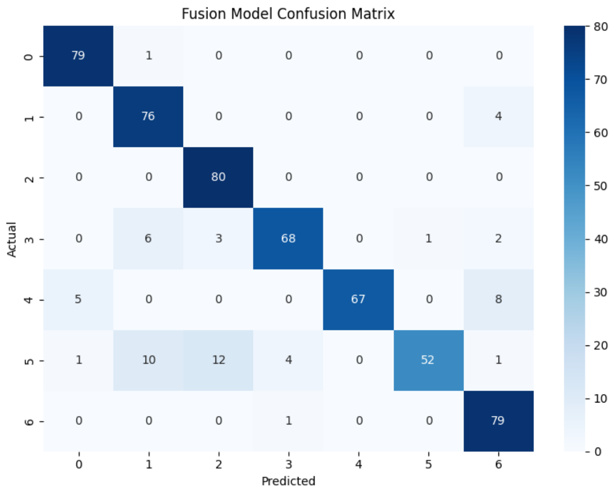
These results are substantially lower than those obtained using speech-based approaches.

This outcome is expected because the TESS dataset contains isolated words such as: Back, chair, young, bar, boat These words do not inherently contain emotional information. Emotional expression in the dataset is conveyed primarily through vocal characteristics rather than lexical content.

As a result, DistilBERT embeddings provide limited discriminative information for emotion classification, causing the model to perform close to random guessing across the seven classes.

5.5 Fusion Model Analysis

Figure 10: Confusion Matrix for Fusion Model



The final multimodal fusion model achieved:

Accuracy = 89.46%
Weighted F1 = 89.16%
The model successfully integrated speech and text representations through feature-level fusion.
Initial experiments involving full fine-tuning of both pretrained encoders resulted in unstable optimization and poor performance. To address this issue, the Wav2Vec2 and DistilBERT encoders were frozen, allowing only the BiLSTM and classifier layers to be trained. This observation suggests that speech already provides highly informative emotional cues, while the textual modality contributes relatively little additional information in the TESS dataset.

5.6 Comparative Discussion

The relative performance of the implemented models can be summarized as:
Wav2Vec2 + BiLSTM > MFCC + CNN > Fusion Model > DistilBERT + BiLSTM
Several observations can be made:

1. Speech Dominates Emotion Recognition

Speech-based models consistently achieved the highest performance. Emotional information such as pitch variation, speaking rate, energy, and intonation is strongly represented in audio recordings, making speech the most informative modality in this dataset.

2. Pretrained Speech Models are Highly Effective

The Wav2Vec2 model significantly benefited from large-scale self-supervised pretraining, enabling it to learn robust acoustic representations that generalize effectively to emotion recognition tasks.

3. Text Alone is Insufficient

The text model demonstrated poor performance because the dataset contains isolated words without contextual information. Transformer language models rely heavily on semantic context, which is largely absent in TESS.

4. Fusion Requires Informative Modalities

Although multimodal fusion is generally expected to improve performance, its effectiveness depends on the quality of each modality.

5.7 Summary of Findings

The experiments demonstrate that speech-based representations are highly effective for emotion recognition on the TESS dataset. The Wav2Vec2 + BiLSTM model achieved perfect classification performance, outperforming both the baseline CNN and the multimodal fusion architecture.
The text-only model performed poorly due to the lack of semantic information in isolated words, highlighting the importance of contextual content for language-based emotion recognition. While multimodal fusion achieved strong results, it did not surpass the best speech-only model because emotional information in TESS is primarily encoded through speech rather than text.

6. Conclusion and Future Work

6.1 Conclusion

This project explored emotion recognition using speech, text, and multimodal fusion approaches on the Toronto Emotional Speech Set (TESS) dataset. A complete machine learning pipeline was developed, including data preparation, feature extraction, model training, and evaluation.

Three main architectures were implemented: an MFCC + CNN baseline model, a Wav2Vec2 + BiLSTM speech model, a DistilBERT + BiLSTM text model, and a multimodal fusion model combining speech and text representations. Experimental results showed that speech is the most informative modality for emotion recognition in the TESS dataset. The Wav2Vec2 + BiLSTM model achieved the best performance with 100% accuracy and weighted F1-score, while the MFCC + CNN baseline also produced excellent results.

The text-based model achieved significantly lower performance because the dataset contains isolated words with limited semantic information. The fusion model achieved strong results after freezing the pretrained encoders, but it did not surpass the performance of the speech-only model. Overall, the project demonstrates the effectiveness of pretrained speech representations for emotion recognition and highlights the importance of selecting appropriate modalities based on dataset characteristics.

6.2 Future Work

Several improvements can be explored to enhance multimodal emotion recognition systems:

1. **Larger and More Diverse Datasets**
   Utilize datasets containing more speakers, languages, and real-world emotional expressions to improve generalization.
2. **Sentence-Level Text Analysis**
   Replace isolated words with complete sentences or conversational transcripts to provide richer semantic information for text-based emotion recognition.
3. **Advanced Fusion Techniques**
   Investigate attention-based and transformer-based fusion methods to better capture interactions between speech and text modalities.
4. **End-to-End Fine-Tuning**
   Fine-tune all model components on larger datasets instead of freezing pretrained encoders during training.
5. **Real-Time Applications**
   Extend the system for deployment in virtual assistants, customer service analytics, mental health monitoring, and human-computer interaction systems.

These improvements can further enhance the robustness, scalability, and practical applicability of multimodal emotion recognition systems

References

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